Simulation of Motor-Enhanced Random Motion

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Abstract (of the original paper)

Intracellular transport is often described in terms of two familiar mechanisms: thermal Brownian motion and directed, motor-driven motion that occurs along cytoskeletal filaments. Over the past two decades, however, experiments have revealed a third, distinct mode of motion. This motion arises when particles are trapped within a dynamic cytoskeletal mesh and undergo displacements that are random in direction, either by hopping between traps or by movements of the traps themselves, which are mediated by molecular motors. Although diffusive in form, this motion occurs at amplitudes far exceeding thermal expectations and dominates intracellular transport for particles larger than individual proteins on biologically relevant timescales.

This phenomenon has been described under many names, including active diffusion, enhanced Brownian motion, ATP-dependent jiggle, and boosted random walk. We review the historical development of this subset of active diffusion, then summarize results of a new approach for analysis of this motion developed by our late colleague Dr. George Holzwarth, based on pattern recognition methods. We also propose a new name for this type of motion that captures its essential characteristics: motor-enhanced random motion.

By integrating experimental evidence, theoretical interpretations, and the results from a simple simulation, we argue that intracellular transport is best understood as a balance among Brownian, driven, and motor-enhanced random motion, with additional cellular processes modulating their relative effectiveness. Recognizing motor-enhanced random motion as a distinct and ubiquitous transport mechanism clarifies long-standing ambiguities in the literature and highlights new directions for investigating how cells harness stochastic, energy-dependent fluctuations to achieve efficient intracellular organization.

Simulation of Driven, Brownian, and Motor-Enhanced Random Motion

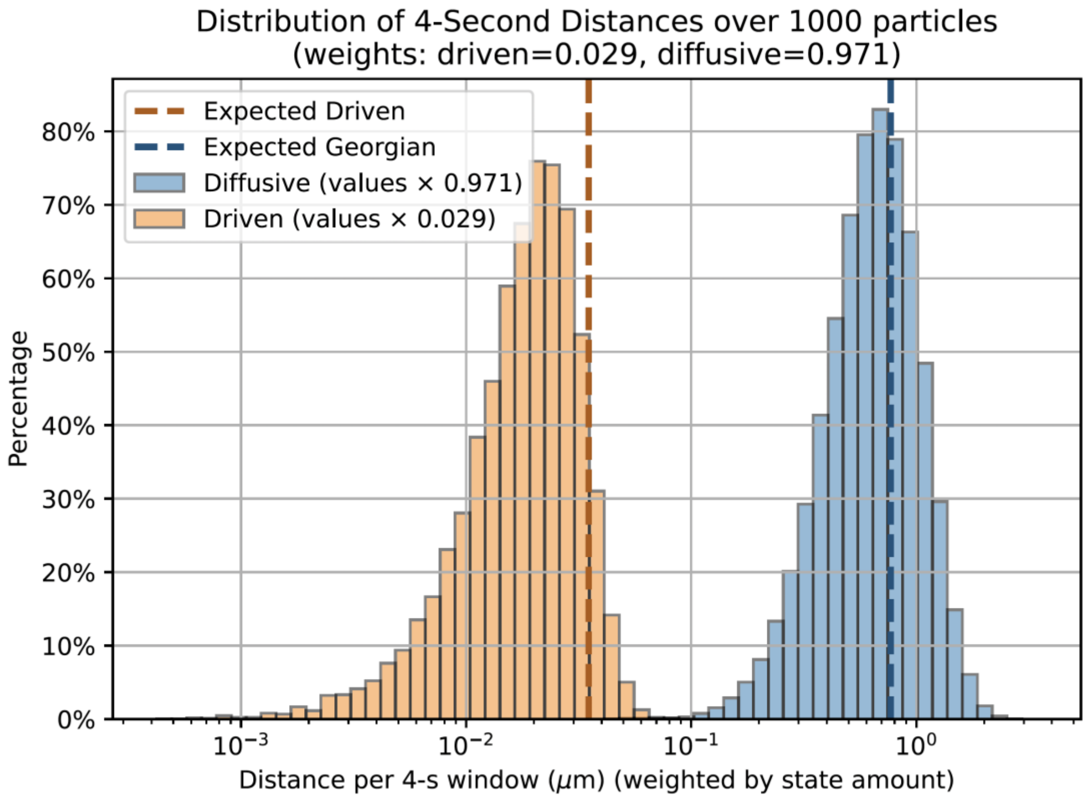
In this section, we describe a Python simulation that extends short-term experimental data and demonstrates that Brownian motion, motor-enhanced random motion, and driven motion are all nontrivially exploited by the cell to transport particles. While there have been numerous experimental studies on motor-enhanced random motion *in vitro* as described previously, these studies are often limited by technological limitations in particle tracking. For example, if fluorescent tags are used to label individual particles, these fluorescent molecules will undergo photobleaching during the tracking process and, as a result, are only effective for a few seconds' worth of data (Holzwarth et al. 2020). Meanwhile, depending on the specific process, intracellular transport takes orders of magnitude longer than what can be measured experimentally. So, while these studies are able to characterize motor-enhanced random motion in the short term, they are not able to reliably describe particle behavior over prolonged time periods.

Here, we present a brief overview of our simulation and some observations of simulated particles over prolonged periods of time. This simulation is not exact and serves as a proof-of-concept for modelling motor-enhanced random motion. More detailed information about the implementation can be found in the Supplementary Information, and the source code can be found at https://github.com/songk42/merm-simulation.

Our simulation is set up to model a mixture of thermal Brownian motion, motor-enhanced random motion, and motor-protein powered driven motion within a set of particles inside the cell. Motor-enhanced random motion was modeled in two parts: the thermal vibration within each trap and the overall hopping between traps (i.e. the "actual" motor-enhanced random motion). Meanwhile, driven motion was modeled as a one-dimensional random walk directed radially from the nucleus, meaning a particle will either move directly toward or away from the nucleus along microtubules. Transitions between each bout of motion (i.e. between motor-enhanced random motion and driven motion, and between distinct traps) were assumed to be instantaneous for the sake of the simulation.

The relative frequencies of each type of motion can be adjusted within the simulation, as well as other physical characteristics of the environment being simulated (e.g. particle and trap size, cytoplasm viscosity, length of bouts of motor-enhanced random motion versus driven motion).

Case study: Vesicular stomatitis virus ribonucleoproteins (VSV RNPs)



Histogram of "fluxes" over a 4-second window as derived from our simulation. As in the experimental data, motor-enhanced random motion contributes about 20 times as much to the overall particle displacement as driven motion does (0.692 μm vs. 0.020 μm). The blue and orange dashed lines indicate experimental values for distances for motor-enhanced random motion and driven motion respectively, as found in (Holzwarth et al. 2020) (0.77μm and 0.035μm).

We demonstrate our simulation by modelling the conditions found in (Holzwarth et al. 2020). We run our simulation on 1000 simulated RNP particles for 6000 seconds each. We model the enclosing cell as a circle with radius 15 μm and an impenetrable nucleus as a concentric circle with radius 5 μm; the cells used in (Holzwarth et al. 2020) are approximately this size. Particles are initialized halfway between the nucleus and the edge of the cell (i.e. 7.5 μm from the center).

It was found experimentally that roughly 3% of sampled tracks over a four second time period should contain some amount of driven motion (Holzwarth et al. 2020). As our simulation is discretized by time steps, we rework this observation as 0.4% of the total time being occupied by driven motion. The derivation of this measure can be found in the Supplementary Information.

First, we analyze each 4-second window of each particle's trajectory as in (Holzwarth et al. 2020). Like in that experiment, we find that motor-enhanced random motion contributes around 20 times as much to the overall particle flux as driven motion does (Fig. 1). “Flux” in this context is as defined in (Holzwarth et al. 2020), i.e. the root mean square displacement from the particle’s initial position.

Then, we examine particle behavior across the entire 6000-second timeframe. Here, we find that driven motion contributes to around 16.4% of the total particle distance traveled (23.63 μm traveled under driven motion vs. 120.16 μm under motor-enhanced random motion). Over this longer timescale, driven motion has a larger impact on overall particle behavior than over the 4-second window, but the majority of the motion is still determined by motor-enhanced random motion. This may be a surprising result, since driven motion is usually considered the major transport mechanism. VSV RNPs show a relatively low proportion of driven motion compared to other particles. For example, for early endosomes in the same cell type, about 22% of states along each particle’s trajectory are spent in driven motion (Moran et al. 2024). In addition, this simulation assumes that driven motion is a perfectly uniform 1-dimensional random walk, with equal probability of moving towards or away from the center of the cell. In reality, this fraction is determined by the particle’s relative binding affinity for kinesin vs. dynein, giving a bias in the particle’s driven motion meaning that driven motion should overall have a larger impact on the overall trajectory than what is shown in our simulation, but still smaller than expected since motor-enhanced random motion takes on part of the transporting role.

This simulation shows that both motor-enhanced random motion and driven motion have a noticeable impact on particle motion in this case. It can reproduce experimental conditions for VSV RNPs exiting the cell and is extensible to other particles and environments, given the right parameters. As this *in silico* model performs a large amount of extrapolation (hours of simulation, as opposed to seconds of experimental reference data), we are not guaranteed to see the same sort of particle behavior *in vitro* or *in vivo* if we were able to measure for that long. Moreover, since the original VSV RNP study by Holzwarth et al. was published, there has been progress on better methods for experimental particle tracking, including longer-lasting tags, novel tagging methods to allow currently-existing labels to last longer when being imaged, and improved microscopy methods (Si et al. 2025; Zhang, Shen, and Yang 2025; Moosmayer et al. 2024). So, we look forward to future studies that experimentally study motor-enhanced random motion on longer time scales.

Bibliography

Holzwarth, George, Arnav Bhandari, Lucas Tommervik, Jed C. Macosko, David A. Ornelles, and Douglas S. Lyles. 2020. “Vesicular Stomatitis Virus Nucleocapsids Diffuse through Cytoplasm by Hopping from Trap to Trap in Random Directions.” *Scientific Reports* 10 (1). Nature Publishing Group: 10643. doi:10.1038/s41598-020-66942-6.

Moosmayer, Thea, Kamila A. Kiszka, Volker Westphal, Jasmin K. Pape, Marcel Leutenegger, Heinz Steffens, Seth G. N. Grant, Steffen J. Sahl, and Stefan W. Hell. 2024. “MINFLUX Fluorescence Nanoscopy in Biological Tissue.” *Proceedings of the National Academy of Sciences* 121 (52): e2422020121. doi:10.1073/pnas.2422020121.

Moran, Steven, Ryan Oglietti, Kathleen C. Smith, J. Macosko, G. Holzwarth, and D. Lyles. 2024. “Mechanisms of Active Diffusion of Vesicular Stomatitis Virus Inclusion Bodies and Cellular Early Endosomes in the Cytoplasm of Mammalian Cells.” *PLOS ONE*. doi:10.1371/journal.pone.0290672.

Si, Dongjuan, Zhihao Dong, Gang Xing, Longfang Yao, Jun Lu, Min Ma, Yue Yuan, et al. 2025. “A Bright and Regenerative Array Tag with Fluorogenic Ligands for Long-Term Live-Cell Nanoscopy and Single-Molecule Tracking.” *Nature Communications* 17 (1): 576. doi:10.1038/s41467-025-67273-8.

Zhang, Lei, Dongwen Shen, and Jiazhen Yang. 2025. “Advancing Single-Molecule Biophysics: Next-Generation Organic Fluorophores with Tailored Labeling Strategies.” *Chemical & Biomedical Imaging* 3 (9): 572–98. doi:10.1021/cbmi.5c00007.

Supplementary Information

Simulation setup

All parameters are selected to match the vesicular stomatitis virus (VSV) ribonucleoprotein (RNP) experiments performed in (Holzwarth et al. 2020).

Simulation of Motor-Enhanced Random Motion with Langevin Equation

MERM was modeled in two parts: the vibration within each trap and the overall hopping between traps (discussed alongside the driven motion setup below).

Each trap was represented as a harmonic potential, where a particle x away from the center of the trap experiences a restoring force of $F(x) = -kx$ and k is the spring constant. Within our simulation k was selected uniformly from the interval $6.0 \cdot 10^{-7} \leq k \leq 2.6 \cdot 10^{-6}$ N/m.

The pseudo-Brownian motion within each trap was modeled with the stochastic Langevin equation for harmonic potentials:

$$m\ddot{x} = -\gamma\dot{x}(t) + kx(t) + W(t)\sqrt{2k_B T\gamma}$$

where m is the mass of the particle, γ is the viscous drag coefficient, k is the spring constant of the trap, k_B is Boltzmann's constant, T is the temperature in Kelvin, and $W(t)$ is a stochastic Wiener process with mean 0 and standard deviation 1. (A Wiener process is a process such that $W(t) - W(t_0)$ is normally distributed with mean 0 and variance $t - t_0$ for any $0 \leq t_0 \leq t$.) The values of these variables used were $\gamma = 9.72 \cdot 10^{-9}$ and $T = 310.15$ K. Within the simulation, the coordinates for each particle were recalculated after every 0.001 simulated second. In other words, this Langevin equation was evaluated using numerical integration in the form

$$x[t] = x[t - \Delta t] - \frac{k(x[t - \Delta t] - x_c)\Delta t}{\gamma} + N(t)\sqrt{2D\Delta t}.$$

$N(t)$ is a normally distributed random number with mean 0 and standard deviation 1, and D is the diffusivity of the cytoplasm. Δt is the interval at which we update the RNPs' coordinates, so here $\Delta t = 0.001$.

The inertia term from above was ignored because it was sufficiently small. Both the x and y coordinates were computed in this manner, independently of each other.

Simulation of Driven Motion

Driven motion was modeled as a one-dimensional random walk directed radially from the nucleus, meaning a particle will either move directly toward or away from the nucleus along microtubules. The step sizes were selected from a Gaussian distribution N(0.001μm, 0.001), assuming the average velocity is 1 μm per second. To match experimental data, bouts of driven motion lasted for a time defined as a Gaussian distribution of mean 0.6 seconds and standard deviation 0.2 seconds. During a particular bout of driven motion, the particle does not change direction. However, each bout has an equal probability of being directed toward or away from the nucleus.

Switching Between Driven and Motor-Enhanced Random Motion States

We switch between driven and MERM states by running each for a time period t, which is sampled from a normal distribution with mean and standard deviation matching experimental observations. Transitions between each bout of motion (i.e. between MERM and driven motion, and between distinct traps) were assumed to be instantaneous for the sake of the simulation.

It was found experimentally that roughly 3% of sampled tracks over a four second time period should contain some amount of driven motion (Holzwarth et al. 2020). As our simulation is discretized by time steps, we rework this observation as 0.4% of the total time being occupied by driven motion. This measure can be derived as follows.

We can approximate the state selection process by thinking of the starts of driven bouts as events in a homogeneous Poisson distribution with a constant rate λ. Then, the probability q of at least one event (i.e. one instance of driven motion) in a window of duration t is equal to $q = 1 - e^{-\lambda(T+\mu)}$ where:

- μ is the mean duration of driven bouts
- T is the observation window length
- q is the probability of at least one driven event in the window
- λ is the rate of bouts

Additionally, the fraction of time spent in the driven state $p_{driv} = 1 - e^{-\lambda\mu}$.

Then, isolating λ we can derive from $q = 1 - e^{-\lambda(T+\mu)}$ that $\lambda = \frac{-\ln(1-q)}{T+\mu}$. Substituting into $p_{driv} = 1 - e^{-\lambda\mu}$, $p_{driv} = 1 - (1-q)^{\frac{\mu}{T+\mu}}$.

We previously defined the q, the probability of driven motion being contained in any window, as 3%. We previously defined μ, the mean duration of driven bouts, as 0.6s. Finally, we have defined T, the observation window length, as 4s.

Thus, $p_{driv} = 1 - (1 - 0.03)^{\frac{0.6}{4.6}} = 0.4\%$. This value was used to clearly sample driven and diffusive lengths within the simulation.

Selecting traps for motor-enhanced random motion Each bout of MERM lasted for a duration sampled from a normal distribution N(0.03 seconds, 0.03), and distances between traps were sampled iteratively from a gamma distribution $\Gamma(\alpha$=1.41, θ=1.4e-8) (in μm) to match experimental data . We did not place restrictions on trap locations as long as they did not enter the nucleus; within a real cell, the cytoskeleton defining these traps would both not be organized into a neatly-defined grid and vary its positions over time.

Simulation Characteristics

We run our simulation on 1000 simulated RNP particles for 6000 seconds each. We model the enclosing cell as a circle with radius 15 μm and an impenetrable nucleus as a concentric circle with radius 5 μm; the cells used in Holzwarth et al., “Vesicular Stomatitis Virus Nucleocapsids Diffuse through Cytoplasm by Hopping from Trap to Trap in Random Directions.” are approximately this size. Particles are initialized halfway between the nucleus and the edge of the cell (i.e. 7.5 μm from the center).

Motor-enhanced random motion trap sizes and distances

In the main body of the Perspective, we analyzed the results of running the full 6000-second track. Here, we compare the simulation’s performance on 4-second tracks and verify that it matches what was observed in (Holzwarth et al. 2020).

One way to characterize our simulation is to examine its physical attributes. For example, we can see from Figure 1 that the distribution of distances between adjacent traps for MERM/trap hopping remains relatively consistent between experimental and simulation data.

As in Holzwarth et al. (2020), we use $\sigma = \frac{\sigma_x + \sigma_y}{2}$ as an indicator of trap size, derived from the most probable 2-dimensional Gaussian for each observed trap. The simulation provides a much narrower range of possible trap sizes than what is experimentally shown (see Figure 2). This could be due to each trap’s parameters for thermal Brownian motion; while in the simulation, each trap is given the same parameters (and a small amount of noise), when measured *in vitro/vivo* particles may be more varied in their motions inside the traps depending on their surrounding conditions.

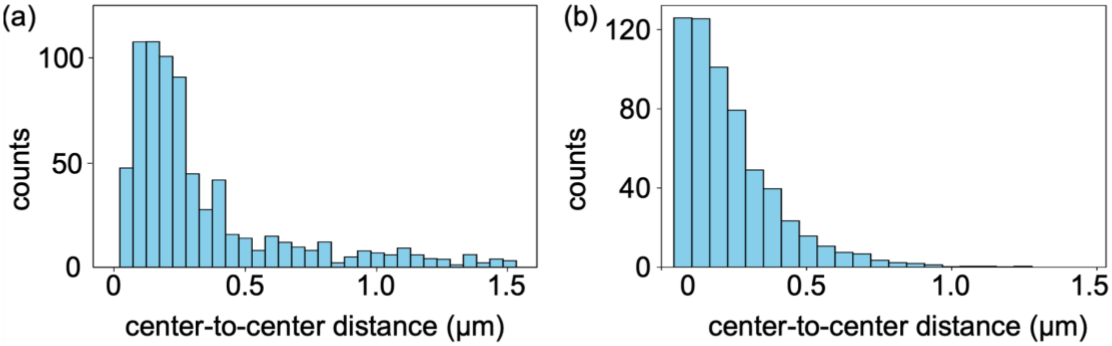


Figure 1: The simulated distance between MERM traps (“center-to-center distance”, subfigure (b)) is very similar to the experimentally determined displacement (subfigure (a)). The simulation has mean trap distance 0.236 μm, while the experimental data peaks around that value as well (Subfigure (a) taken from Holzwarth et al., 2020.)

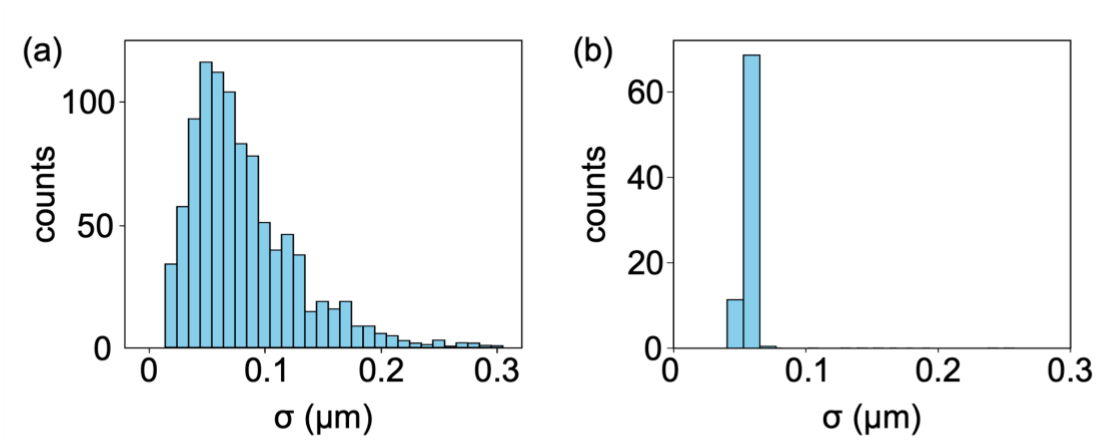


Figure 2: The MERM trap sizes generated by the simulation (subfigure (b)) follow a much narrower distribution as those determined experimentally (subfigure (a)), but with comparable mean: 0.0562 μm, as opposed the experimental value of 0.0825 μm. (Subfigure (a) taken again from Holzwarth et al., 2020.)